

Workshop

Building and Accessing Cortex Agents with Google Cloud & Snowflake

Task 0. Setup and Requirements

Before you click the Start Lab button

Read these instructions. Labs are timed and you cannot pause them. The timer, which starts when you click **Start Lab**, shows how long Google Cloud resources will be made available to you.

This hands-on lab lets you do the lab activities yourself in a real cloud environment, not in a simulation or demo environment. It does so by giving you new, temporary credentials that you use to sign in and access Google Cloud for the duration of the lab.

What you need

To complete this lab, you need:

- Laptop and access to a standard internet browser (Chrome browser recommended).
- Time to complete the lab.

Note: If you already have your own personal Google Cloud account or project, **do not use it** for this lab.

Note: If you are using a Chrome OS device, open an Incognito window to run this lab.

Alternatively, you can log out of all other Google / Gmail accounts before beginning the labs.

How to start your lab and sign in to the Google Cloud Console

Tip: When prompted, select the option to use a new Chrome profile to keep this lab separate from other Chrome activities.

1. Click the **Start Lab** button. On the left is a panel populated with the temporary credentials that you must use for this lab.

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked.
[Learn more.](#)

Open Google Cloud console

Username

student-04-8b08d06a5b0i 

Password

ZYWNfhEx5l7o 

Project ID

qwiklabs-gcp-03-697645i 

Region

us-central1 

- Copy the username, and then *right-click* **Open Google Console** and select **Open Link in Incognito Window**. The lab spins up resources, and then opens the incognito window. You'll be prompted to read and acknowledge the Welcome to Your New Account details. Click "I Understand".

You'll be presented with the Terms and Conditions. Verify that your account matches the Username from your lab credentials. Click the checkbox to agree to the Terms of Service, and then click on "Agree and Continue".

Important: You must use the credentials provided for the lab. Do not use your own Google Cloud Training credentials. If you have your own Google Cloud account, do not use it for this lab (avoids incurring charges).

Google Cloud

Welcome!

Create and manage your Google Cloud instances, disks, networks, and other resources in one place.

 student-04-8b08d06a5b0a@qwiklabs.net [Switch account](#)

Country

United States

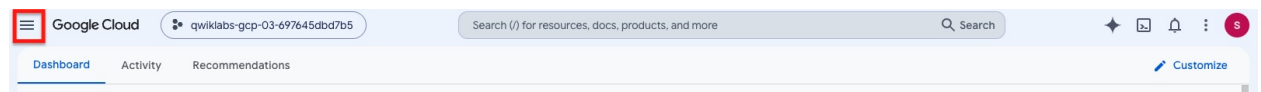
Terms of Service

☒ I agree to the [Google Cloud Platform Terms of Service](#), and the terms of service of [any applicable services and APIs](#). If I am using a Starter Tier project, the [Starter Tier Terms of Service](#) apply to my use of all resources in that project.

[Agree and continue](#)

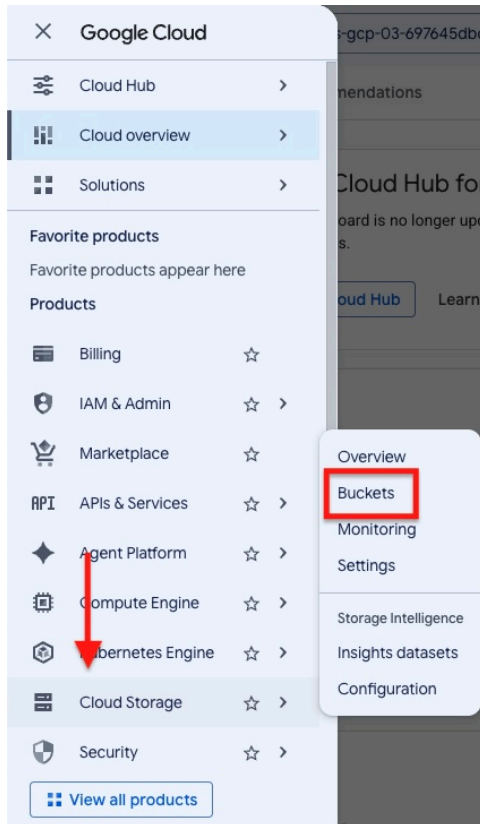
- After a few moments, the Google Cloud will open.

Note: You can view the menu with a list of available Google Cloud Products and Services by clicking the **Hamburger menu** at the top-left.

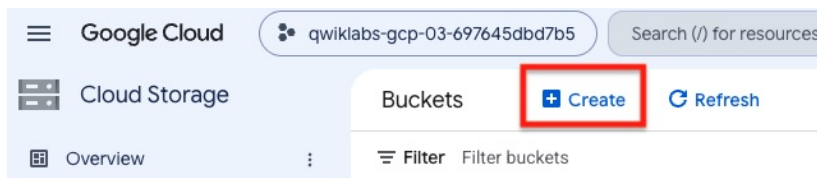


Task 1. Setup Your Google Cloud Storage Bucket

- We will create a Google Cloud Storage bucket to hold the data for the Iceberg table that we will be creating in Snowflake. In the left navigation panel, hover over **Cloud Storage** and select **Buckets**.

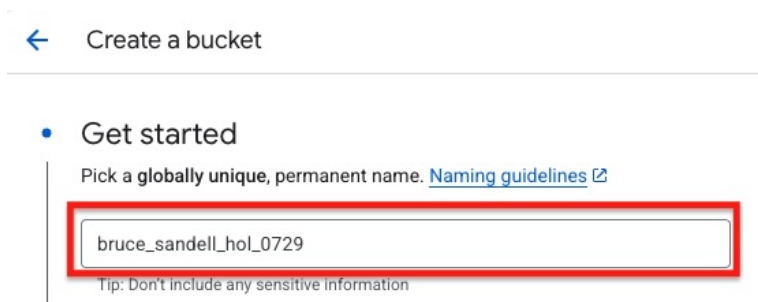


2. Click the "Create" link to start the bucket creation process.



3. Add your bucket name as *firstname_lastname_hol_0729*. (Note, keep this bucket name copied and pasted into a notepad/textpad as we will need it later.)

We'll be using all of the default values for creating the bucket, so you can scroll all the way to the bottom of the page and click "Create".



4. Leave **Enforce public prevention** checked and click **Confirm**.

Public access will be prevented

This bucket is set to prevent exposure of its data on the public internet.

Keep this setting enabled unless you have a use case that requires public access (such as static website hosting). You can change it now or later. [Learn more](#)

☒ Enforce public access prevention on this bucket 

☐ Don't show this message again

Cancel **Confirm**

5. **Return to your Snowflake Notebook.** Once you have completed the step where you CREATE EXTERNAL VOLUME, return here with the value for STORAGE_GCP_SERVICE_ACCOUNT as described in the Notebook.
6. Click on the **Permissions** tab on the bucket screen so we can add permissions for the Snowflake service account. Then click the **Grant Access** link.

← Bucket details

📁 bruce_sandell_hol_0729

Location	Storage class	Public access
us (multiple regions in United States)	Standard	Not public

Objects Configuration **Permissions** Protection Lif

Public access

Not public

This bucket is not publicly accessible since public access is being prevented. Because of this restriction, objects cannot be publicly shared over the internet. [Learn more](#)

Principals restricted from bucket access:
allUsers, allAuthenticatedUsers

[Remove Public Access Prevention](#)

Permissions

[View by principals](#) View by roles

 [Grant access](#) [Remove access](#)

- On the **Grant Access** screen, paste the Snowflake STORAGE_GCP_SERVICE_ACCOUNT (which would have been returned when you ran the SQL command “DESC EXTERNAL VOLUME hol_gcs_vol;” in Snowsight) into the **New principals** box, and click enter. Under **Assigned roles**, click inside to open the filter box. In the filter enter **storage admin**, and select the **Storage Admin** entry. Then click **Save**.

Grant access to "bruce_sandell_hol_0729"

Grant principals access to this resource and add roles to specify what actions the principals can take. Optionally, add conditions to grant access to principals only when a specific criteria is met. [Learn more about IAM conditions](#)

Resource

bruce_sandell_hol_0729

Add principals

Principals are users, groups, domains, or service accounts. [Learn more about principals in IAM](#)

New principals *

k86m30000@gcpuscentral1-1dfa.iam.gserviceaccount.com
X
?

Assign roles

Roles are composed of sets of permissions and determine what the principal can do with this resource. [Learn more](#)

Role *

Storage Admin
▼

IAM condition (optional) ?

+ Add IAM condition

Grants full control of buckets and objects.

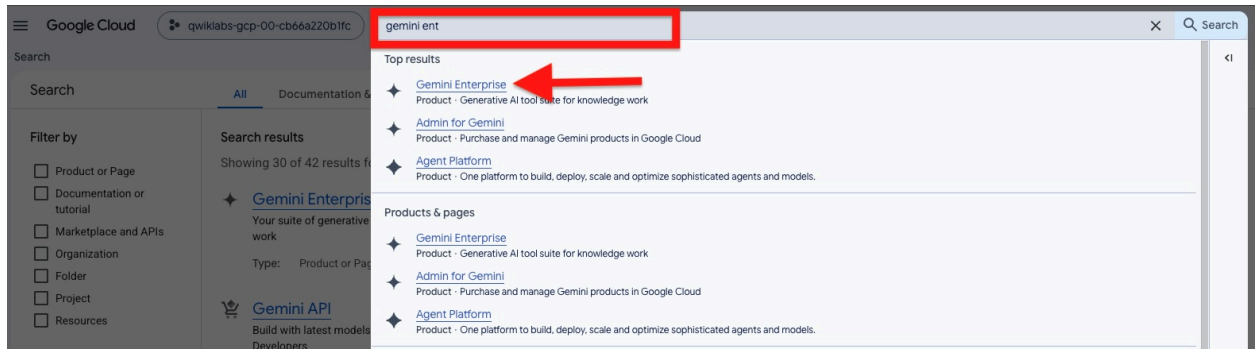
+ Add another role

Save

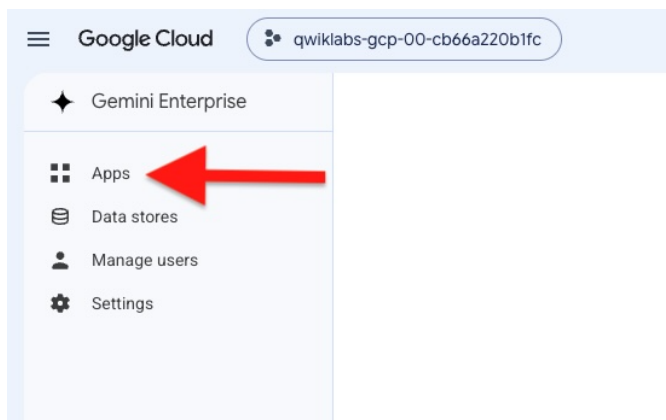
Cancel

Task 2. Create your Custom MCP Server Data Source

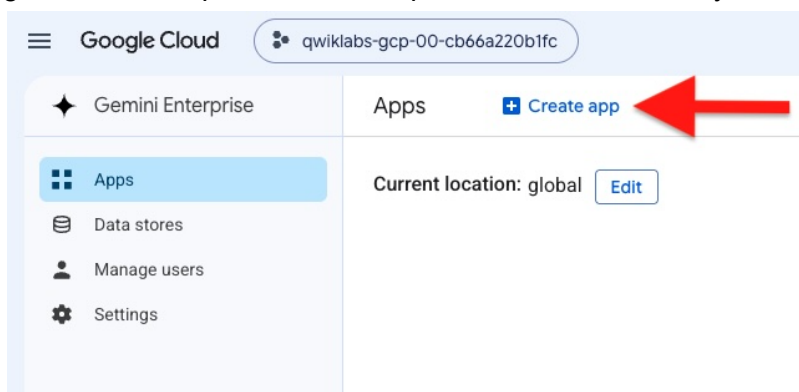
- Before starting this step, please ensure that you have completed the creation of your MCP Server in the Snowflake Lab instructions.
- In the Search Field at the top of the Google Cloud Console, enter **Gemini Enterprise** and then click on Gemini Enterprise from the list of products displayed to go to the Gemini Enterprise Home Screen.



- From the Gemini Enterprise Home Screen, Click on **Apps** in the left-side panel.



- Select "Create App". A Gemini Enterprise App is a centralized, AI-powered workplace platform designed for teams to collaborate, automate workflows, and deploy custom AI agents without writing code. It acts as a unified hub where employees can access grounded enterprise data and specialized tools securely.



- Choose a name for your Gemini Enterprise App. You can use the one suggested in the Notebook, or create your own. Press "Create" to create your new app. It will take a few seconds to create the new app.

Google Cloud qwiklabs-gcp-00-cb66a220b1fc Search (/) for resources, docs, prod

Gemini Enterprise Create

Choose a display name

App name *
hol_cortex_gemini_economics_agent
ID: hol-cortex-gemini-economic_1784697273902. It cannot be changed later. [Edit](#)

Choose a location

Multi-region *
global (Global)

If you do not have compliance or regulatory reasons to locate your data in a particular multi-region, we recommend you choose the global location. You cannot change this later. For important information about multi-regions, see [Gemini Enterprise locations](#)

Advanced options

Create Cancel

6. From the Apps home page, click on “Connected data stores”

Google Cloud qwiklabs-gcp-00-cb66a220b1fc Search (/) for resources, docs, products, and more

Gemini Enterprise Apps > hol_cortex_gemini_economics_agent

Dashboard

Connected data stores

Welcome, student!

Your Gemini Enterprise webapp is ready

Users with appropriate credentials can securely sign in via this URL

https://vertexaisearch.cloud.google.com/home/cid/e94cc81c-ac94-481a-933f-4327c7abe4d6?hl=en_US

Copy URL Configure webapp

7. Click on “New Data Store”

Connected data stores + New data store Add existing data stores

Filter Enter property name or value

Name	Type	Status	Last data sync	Last update	Date created
No data stores are connected to this Gemini Enterprise app yet, click 'Add existing data store' to connect an existing one, or 'New data store' to create a new one and automatically connect it to the app					

8. Scroll through the list of available data stores until you get to the “MCP Servers” section. In the data source named “Custom MCP Server”, click on “Add MCP server”.

Third-party data sources [Show all →](#)

Third-party data sources that you can connect to your instance.



Microsoft Entra ID
Import data from your Microsoft Entra ID site.

[Add data source](#)



OneDrive
Import data from your OneDrive site.

[Add data source](#)



Microsoft Outlook
Import data from your Outlook site.

[Add data source](#)

MCP servers [Show all →](#)

MCP servers from your project's configured Agent Gateway registries, or a custom MCP server.

 Configure an Agent Gateway to discover and use MCP servers from your project's Agent Gateway registries. [Configure Agent Gateway](#)



Custom MCP Server
Connect your own Model Context Protocol (MCP) server to use custom tools.

[Add MCP server](#)

9. Leave OAuth 2.0 selected as the verification method. Populate the fields with the data supplied by the Snowflake Notebook. Every field should be populated except "Authorization URL Parameters".

MCP Server Configuration

Provide your MCP server URL and authentication details.

Note when you configure, and allow your End Users to access, a Third-Party (3P) MCP server, their query string and token are subject to the 3P's Terms of Service and Privacy Policy.

1 Authentication settings

MCP Server URL *

The base URL of your hosted MCP server (e.g., https://mcp.example.com).

Authorization URL *

The OAuth 2.0 authorization endpoint URL.

Authorization URL Parameters

Additional parameters required by the Authorization URL, if any. Format: &key1=value1&key2=value2

Token URL *

The OAuth 2.0 token exchange endpoint URL.

Client ID *

The Client ID for your MCP server's OAuth application.

☐ **Enable PKCE Support**
Enable PKCE (RFC 7636) for additional OAuth security. Recommended if your MCP server's OAuth provider supports PKCE.

Client Secret

The Client Secret for your MCP server's OAuth application.

Scopes

Space-separated list of OAuth scopes.

Verify Auth

Click “Verify Auth” to go through the Snowflake OAuth process and ensure that all of your authentication details are set up correctly. Use the same Username and password that you used when you initially logged into Snowflake (username = USER, Password =). If you are prompted to allow access to your account, click “Allow”.

10. After successfully authenticating, click on “Continue”.

✔ Successfully logged in.

Continue

11. In the “Advanced Options” section, populate the MCP Server Description and MCP Agent instructions. Suggested values for these fields should have been supplied in your

Snowflake Notebook. Click “Continue”.

MCP Server Configuration
Provide your MCP server URL and authentication details.

1 Note when you configure, and allow your End Users to access, a Third-Party (3P) MCP server, identity. Data in such 3P systems is subject to the 3P's Terms of Service and Privacy Policy.

2 Authentication settings

2 Advanced options

MCP Server Description
Snowflake Cortex Agent for US economic indicators (CPI, mortgage rates, unemployment, income)

MCP Agent Instructions
Use the hol-economics-agent tool to answer questions about US economic data including inflation, mortgage rates, unemployment by state, and

Back

Continue Cancel

12. Provide a name for your new data connector. You can use the one supplied by the Snowflake Notebook, or make up your own name. Leave the other fields with their default values. Click “Create”.

Configure your data connector
Configure additional settings for your data connector

Location of your data connector
Multi-region
global (Global)

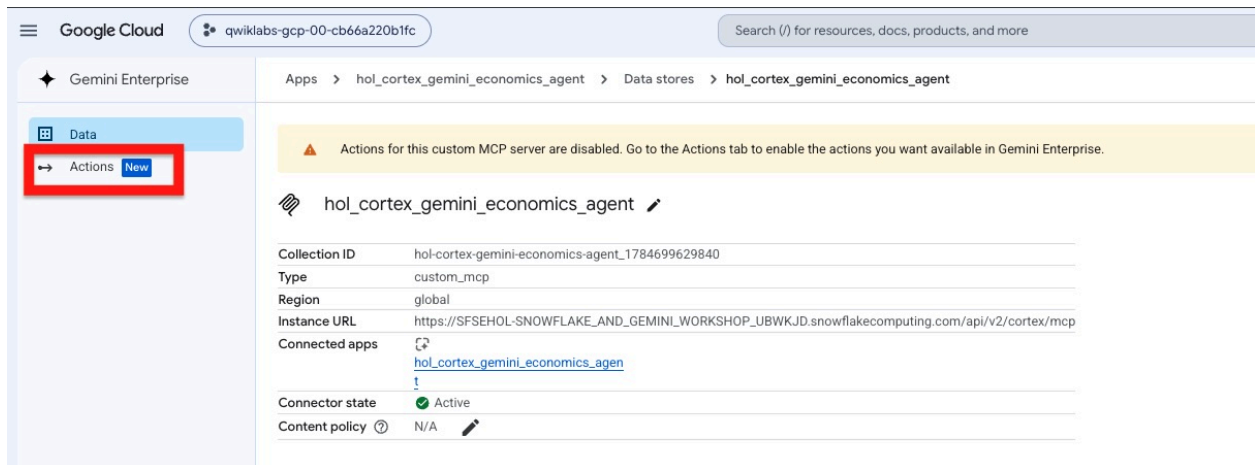
Your data connector name
Data connector name
hol_cortex_gemini_economics_agent

Sensitive data protection policy
Provide a full resource name of a Sensitive Data Protection policy.
Sensitive data protection policy
Format: projects/{project}/locations/{location}/contentPolicies/{policy}

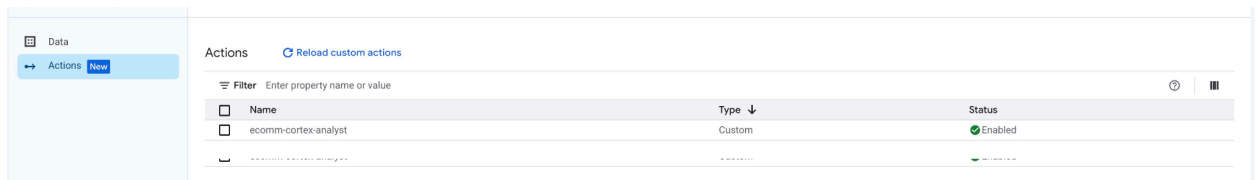
Google Identity will be used for the IDP wide access. No granular access control is supported. [Learn more.](#)

Create Cancel

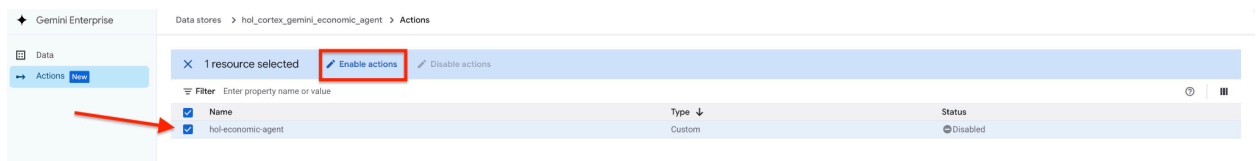
13. It may take several seconds for the data source to be created. Once the data source has been created, you will be returned to the Home screen for your new data source. In the menu on the left, click “Actions”.



14. MCP Servers may expose one or more available actions. Click “Reload custom actions” to get a list of all of the available actions. You will be prompted to go through the OAuth process again. Use the same Username and Password that you used for the earlier authentication step. For this lab, there should only be one available action.



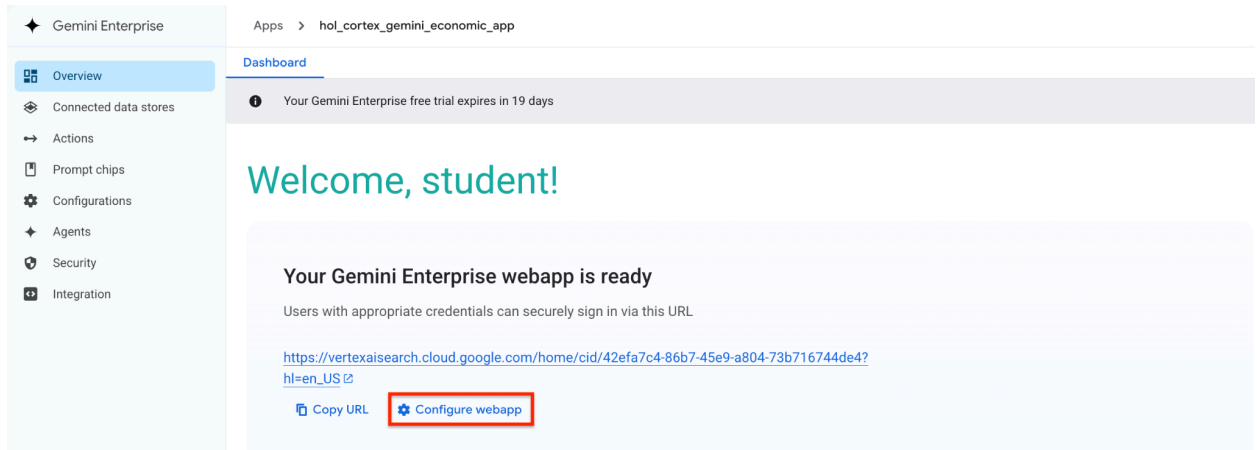
15. You now have the option to select which of the available actions that you want to expose to your end users. Since there is only one action available, we want to enable it. Click on the checkbox next to “hol_economic_agent” and then click on “Enable Actions”



The “Disabled” Indicator should change to “Enabled”.

Task 3. Configure the Gemini Enterprise Webapp

1. Your new app should be ready to customize. Before users start accessing the webapp, you can configure what features will be exposed to them. Click on “Configure webapp” to see all of the configuration options.



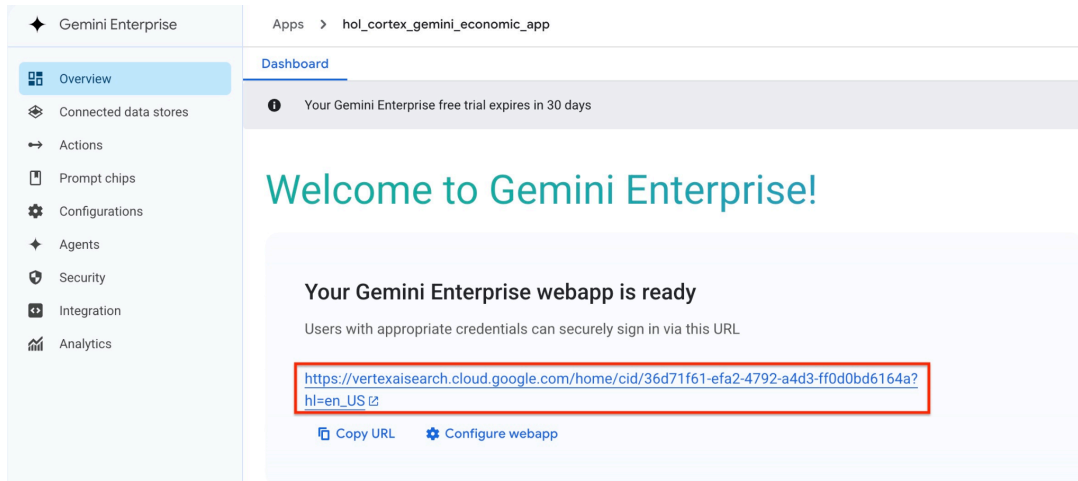
2. Review the features that are available to configure by clicking through the tabs at the top of the page. For this lab, there are two features that we are going to enable. Click on the “Feature Management” tab and enable the features:
Enable agent Designer - allows users to create their own agents using graphical tools
Enable model selector - allows users to pick the most appropriate model for their agent.

The screenshot shows the 'Feature Management' tab in the Gemini Enterprise configuration interface. The left sidebar contains a menu with options: Overview, Connected data stores, Actions, Prompt chips, Configurations (highlighted), Agents, Security, and Integration. Below this, there are links for NotebookLM, CodeAssist, and Manage subscriptions. The main content area is titled 'End user features control' and includes a note about other features being controlled in the Search UI page. A warning message states: 'Please review the Gemini Enterprise data residency commitments page to ensure these features meet your requirements.' The feature list includes: 'Enable agent gallery' (checked), 'Enable agent designer' (checked, with a red arrow pointing to it), 'Enable prompt gallery' (checked), 'Enable model selector' (checked, with a red arrow pointing to it), 'Enable NotebookLM' (checked), and 'Enable session sharing' (unchecked). At the bottom, there are 'Save' and 'Cancel' buttons, with the 'Save' button highlighted by a red box.

3. Navigate back to the app Home page by clicking the “Overview” tab on the left side menu.

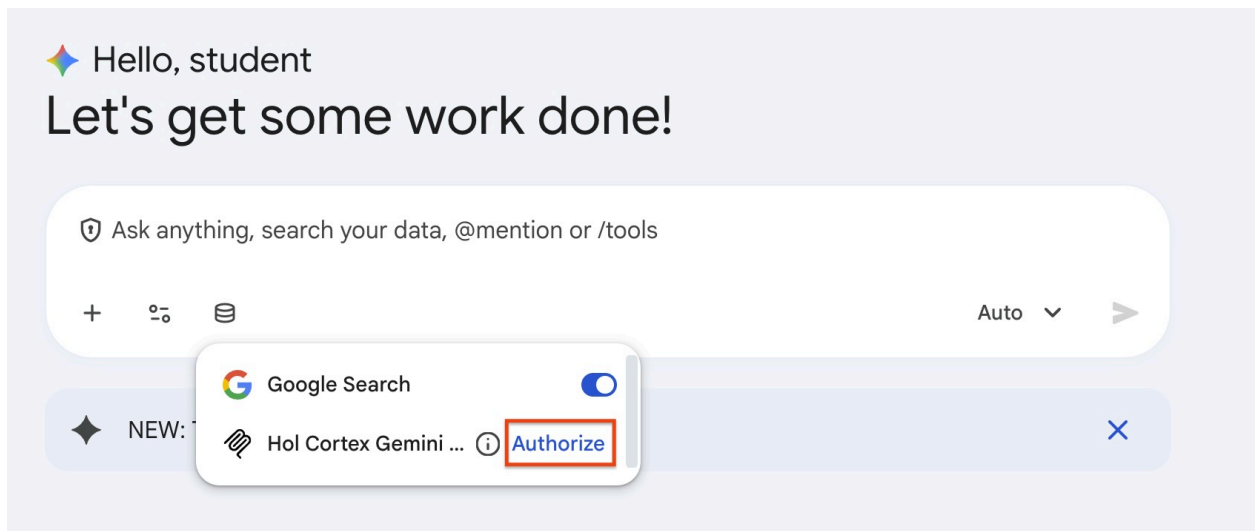
Task 4. Work with your new Gemini Enterprise Webapp

1. Now that the Webapp has been configured, it's time to try it out. Open the Webapp by clicking the link on the App Overview screen.



2. Once the Webapp is open, take a moment to click through the options on the bottom of the “Ask Anything” data entry field. You can add files for searching, enable tools and specify which connectors to use.
3. Click on the disk icon in the data entry field and click “Authorize” to enable the connection to the Snowflake MCP server. You’ll need to authenticate to Snowflake using the same credentials that you have used in previous steps.

Once the authentication has completed successfully, disable Google Search, so that the only enabled data source is your Snowflake MCP Server.



4. Enter a question into the data entry form. A good place to start is: “What kind of data do you have access to?”

Before the question is submitted, you may be requested to accept the prompt that

Gemini is planning to send to your Snowflake agent. You can accept the prompt by clicking on the blue check mark button. You are also free to rate the quality of the generated prompt using the “Thumbs Up” or “Thumbs Down” icon, which will help the Gemini model learn. You can modify the prompt before it is sent as well.

The screenshot shows the Gemini Enterprise interface. At the top, it says "Gemini Enterprise" with a "Plus" button. Below this, there's a "Data access types" section. A prompt input field contains the text "What kinds of data do you have access to?". Below the input field, there's a red "X" button and a blue checkmark button. A red arrow points to the blue checkmark button. At the bottom of the input field, there's a "Text*" label and a placeholder text "What economic indicators and datasets do you have access to?". Below the input field, there are three icons: a thumbs up, a thumbs down, and a menu icon.

5. You will be prompted with some suggested questions, or you can come up with your own. You can have a multi-turn conversation with the data, meaning that the context of the previous question is used as the input to the followup question. For example:

- What was the annual unemployment rate in the United States for 2025?
- Can you break that down by month?
- Can you break down the January data by state?

When you are ready to move on to a new topic, click the “New Chat” link on the left menu. Each time that you open a new chat link, Google Search will be enabled by default, so you may want to disable it.

6. Continue exploring the data!

